Density

Answer key with worked solutions

A. The idea

1. Definition.

> Density is the **mass per unit volume** of a substance: $\rho = \dfrac{m}{V}$. Common units: **g/cm³** (solids), **g/mL** (liquids) and the SI unit **kg/m³**.

B. Calculating density

2. Densities.

> (a) $\rho = \dfrac{96.0\ \text{g}}{12.0\ \text{cm}^3} = \mathbf{8.00\ g/cm^3}$
>
> (b) Convert first: 2.50 kg = 2500 g and 0.500 L = 500. mL.
>
> $\rho = \dfrac{2500\ \text{g}}{500.\ \text{mL}} = \mathbf{5.00\ g/mL}$

3. Metal cube.

> (a) $V = (3.00\ \text{cm})^3 = 27.0\ \text{cm}^3$, so $\rho = \dfrac{241\ \text{g}}{27.0\ \text{cm}^3} = \mathbf{8.93\ g/cm^3}$
>
> (b) Closest value: **copper** (8.96 g/cm³).
>
> (c)
>
> $$\text{error} = \frac{8.926\ \text{g/cm}^3 - 8.96\ \text{g/cm}^3}{8.96\ \text{g/cm}^3} \times 100\%$$
> $$= \mathbf{-0.4\%}$$

4. Displacement.

> $V = 45.2\ \text{mL} - 30.0\ \text{mL} = \mathbf{15.2\ mL} = 15.2\ \text{cm}^3$
>
> $\rho = \dfrac{68.4\ \text{g}}{15.2\ \text{cm}^3} = \mathbf{4.50\ g/cm^3}$, closest to **titanium** (4.51 g/cm³).

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C. Using density

5. Layers.

> (a) From the top: **oil** (0.92 g/cm³), **water** (0.998 g/cm³), **honey** (1.42 g/cm³) at the bottom.
> (b) The bead (1.10 g/cm³) is denser than water but less dense than honey, so it sinks through the oil and water and **rests on top of the honey layer**.

6. Mass and volume.

> (a) $m = \rho V = 0.789\ \frac{\text{g}}{\cancel{\text{mL}}} \times 200.\ \cancel{\text{mL}} = 157.8\ \text{g} = \mathbf{158\ g}$
>
> (b) 1.00 kg = 1000 g, so $V = \dfrac{m}{\rho} = \dfrac{1000\ \text{g}}{11.3\ \text{g/cm}^3} = \mathbf{88.5\ cm^3}$

D. Replicate measurements

7. Five replicates.

> (a) Mean:
>
> $$\bar{\rho} = \frac{(2.68 + 2.71 + 2.69 + 2.72 + 2.70)\ \text{g/cm}^3}{5} = \frac{13.50\ \text{g/cm}^3}{5} = 2.70\ \text{g/cm}^3$$
>
> Deviations: −0.02, +0.01, −0.01, +0.02 and 0.00 g/cm³; the sum of their squares is $1.0 \times 10^{-3}\ (\text{g/cm}^3)^2$.
>
> $$s = \sqrt{\frac{1.0 \times 10^{-3}\ (\text{g/cm}^3)^2}{5 - 1}} = 0.016\ \text{g/cm}^3$$
>
> (b) **2.70 ± 0.02 g/cm³ (n = 5)**
>
> (c) The mean equals the accepted value for aluminium (2.70 g/cm³), so the result is **accurate**; the standard deviation is small (under 1 % of the mean), so it is **precise**.

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E. Challenge

8. The crown.

(a) 1.50 kg = 1500 g and 93.0 mL = 93.0 cm^3:

$$\rho = \frac{1500\ \text{g}}{93.0\ \text{cm}^3} = \mathbf{16.1\ g/cm^3}$$

$$16.1\ \frac{\cancel{\text{g}}}{\cancel{\text{cm}^3}} \times \frac{1\ \text{kg}}{1000\ \cancel{\text{g}}} \times \frac{10^6\ \cancel{\text{cm}^3}}{1\ \text{m}^3}$$

$$= \mathbf{1.61 \times 10^4\ kg/m^3}$$

(b) **No.** Pure gold is 19.3 g/cm^3. The crown's lower density shows it contains a less dense metal, such as silver or copper, mixed with the gold.

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